Vesel

Sets

- $N = \{1, 2, 3, 4\}$: nodes (node 1 is the depot)
- $A = \{(i, j) \in N \times N : i \neq j\}$: arcs
- $P = \{1, 2\}$: products
- $T = \{\text{LO}, \text{HI}\}$: truck types

Parameters

- m_{ij}: distance from node i to node j

$$M = \begin{pmatrix} 0 & 5 & 3 & 3 \\ 5 & 0 & 3 & 7 \\ 3 & 3 & 0 & 5 \\ 3 & 7 & 5 & 0 \end{pmatrix}$$

- q_t: capacity of a truck of type t

$$Q = \begin{pmatrix} 10 \\ 18 \end{pmatrix}$$

- c_t: cost per unit distance of a truck of type t

$$C = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$$

- d_{np}: demand of node n for product p

$$D = \begin{pmatrix} 0 & 0 \\ 2 & 8 \\ 4 & 4 \\ 8 & 2 \end{pmatrix}$$

Decision variables

- $x_{ijp} \geq 0$: amount of product p shipped on arc (i, j)
- $y_{ijt} \in \mathbb{Z}_{\geq 0}$: number of trucks of type t used on arc (i, j)

Model

$$\begin{aligned}
\min \quad & \sum_{(i,j) \in A} \sum_{t \in T} c_t m_{ij} y_{ijt} \\
\text{s.t.} \quad & \sum_{i:(i,j) \in A} x_{ijp} - \sum_{k:(j,k) \in A} x_{jkp} = d_{jp} && \forall j \in N \setminus \{1\}, p \in P \\
& \sum_{p \in P} x_{ijp} \leq \sum_{t \in T} q_t y_{ijt} && \forall (i, j) \in A \\
& x_{ijp} \geq 0 && \forall (i, j) \in A, p \in P \\
& y_{ijt} \in \mathbb{Z}_{\geq 0} && \forall (i, j) \in A, t \in T
\end{aligned}$$